

```
Rotating Device ID = Rotation Counter || Crypto\_KDF(
    inputKey := Persistent Unique ID,
    salt:= Rotation Counter,
    info := "RotatingDeviceID",
    len := 128)
```

(where || is the concatenation operation)

The rotation counter is encoded as 2 bytes using little-endian encoding in the above algorithm, everywhere it appears.

The Rotating Device ID is the concatenation of the current rotation counter and the 16 bytes of the [Crypto\\_KDF](#) result.

5.4.2.4.4.2. TLV Example

Extended data containing just a Rotating Device Identifier would be encoded as the following bytes:

| Offset | Data       | Comments                                                                    |
|--------|------------|-----------------------------------------------------------------------------|
| 0x00   | 0x15       | Control byte for structure with anonymous tag                               |
| 0x01   | 0x30       | Control byte for octet string with 1-byte length and a context-specific tag |
| 0x02   | 0x00       | Context-specific tag for Rotating Device Identifier                         |
| 0x03   | 0x12       | Length of Rotating Device Identifier (e.g. 18 bytes)                        |
| 0x04   | 0xXX..0xXX | Rotating Device Identifier                                                  |
| 0x16   | 0x18       | End of container                                                            |

5.4.2.5. Using BLE

This section provides details of how a device announces its commissionable status or how a device requests fixes to the operational network configuration using BLE technology (see [Bluetooth® Core Specification 4.2 \(amended\)](#) and [Bluetooth® Core Specification Supplement 12](#)). As required in [Section 5.4.2.2, “Announcement Commencement”](#), Nodes currently commissioned into one or more fabrics or already connected to an IP-bearing network SHALL NOT employ this method for commissioning.

5.4.2.5.1. Device Role

Commissionable devices SHALL implement the role of a Generic Access Profile (GAP) Peripheral.

5.4.2.5.2. Channels

There are three advertising channels used by BLE. All three channels SHOULD be used by commissionable devices for BLE advertising.